

ARTICLE INFORMATION

Article title

BD-GI-Desserts13: A Comprehensive Dataset for Recognition and Analysis of GI-Certified Desserts

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Keywords

Bangladeshi desserts; Geographical indication; Food image dataset; Computer vision; Deep learning; Cultural heritage preservation; Image classification; Object recognition

Abstract

BD-GI-Desserts13 is a new visual dataset of 11,692 RGB images (512×512 pixels) of thirteen dessert items certified under the GI relation to Bangladesh collected from verified sites across various districts [1]. Images were collected using consumer smartphones under unconstrained settings across sweet shops, markets and homes and standardized to a fixed resolution for reproducible machine-vision experiments. Each image is annotated with one of thirteen dessert classes including Bogura doi, Cumilla rashmalai, Tangail chomchom, Gopalganj rosogolla, and Kushtia tiler khaza, among others and the dataset is divided into pre-defined training, validation and test subsets using class-stratified sampling 70/15/15 (train/val/test). As well as with key labels, the repository provides metadata on collection district, capture environment, and device information that will allow origin-aware and domain-shift analyses. To demonstrate the reuse potential of the dataset, a lightweight Custom Convolutional Neural Network (CCNN) trained from scratch was implemented as a classification baseline. The model achieved a test accuracy of 99.37%, establishing BD-GI-Desserts13 as a rigorous benchmark, and this fact makes BD-GI-Desserts13 a

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difficult and informative fine-grained dessert recognition test. This result illustrates the potential of the dataset in the construction of strong models and its possibility to further the research in food image recognition and cultural heritage conservation.

SPECIFICATIONS TABLE

Subject	Computer Science – Artificial Intelligence; Computer Vision
Specific subject area	Image-based recognition and analysis of Geographical Indication (GI) certified Bangladeshi desserts using Convolutional Neural Networks (CNNs)
Type of data	Image
Data collection	Photos were shot on location of real GI dessert producers with naming conventions used by the districts and desserts (Cumilla Rashmalai, Tangail Chomchom, Sherpur Chanapayesh etc.), and were photographed in different positions and distances, processed according to quality, anonymized to exclude recognizable faces, and sorted by dessert type with related information.
Data source location	Geographical Indication registered districts across Bangladesh, including Bogura (doi), Brahmanbaria (Chhanamukhi Mishti), Cumilla (Rashmalai), Gopalganj (Rosogolla), Muktagacha in Mymensingh (Monda), Natore (Kacagolla), Netrokona (Balish Mishti), Sherpur (Chanapayesh), Tangail (Chomchom and Sondesh), and Kushtia (Tiler Khaza).
Data accessibility	Repository name: Mendeley Data Data identification number: DOI: 10.17632/5n328km3rr.1 Direct URL to data: https://data.mendeley.com/datasets/5n328km3rr/1 The dataset is publicly accessible without registration or login. Visit the URL above, which directs to the Mendeley Data repository page for BD-GI-Desserts13 (Version 1, published 8 April 2026). Click <i>Download All</i> to obtain the complete dataset as a single ZIP archive, or browse and download individual class folders directly from the file browser on the repository page.
Related research article	None

1 VALUE OF THE DATA

- Bangladeshi desserts like Bogura doi, Muktagacha monda, Cumilla rashmalai, and Tangail chomchom are legally protected GI products whose identity is closely linked to where they are made. However, previous visual datasets either don't mention GI status or mix regions, which makes it hard to do origin-aware analysis. BD-GI-Desserts13 directly fills this gap by only looking at GI desserts that have been

officially certified and have metadata about where they came from at the district level.

- The dataset enables study of fine-grained visual recognition of food classes which are location-specific. The disparities between classes are minimal and disparities within classes are related to the manner in which the food is served, the lighting and the manner in which the shop operates. Thus, it can be used to compare both heavy-duty and light-duty architectures and their usage in real world applications.
- Fixed class-stratified divisions to train, validate and test on, and the amount of images per class are specified, thereby enabling the comparisons of machine-learning and deep-learning models in a repeatable fashion. This aligns with other dataset articles on agriculture and food recognition where obvious divisions and statistics are provided to aid in model development. Metadata will include the GI district, the place of the capture of the food (indoor shop, an outdoor stall, a home kitchen), and the smartphone model. This can be done to investigate domain shift, cross-device generalisation, and environmental variation resistance in ways that goes beyond traditional single-domain food datasets.
- Mobile apps to tourists and shoppers, automated GI authenticity verification at retail counters, and educational resources can be developed using the dataset, displaying pictures and text of Bangladeshi GI desserts. This links computer vision studies with the conservation of cultural heritage and the enhancement of the local economy.

2 BACKGROUND

Bangladesh has also begun to include in its official list of Geographical Indication products, a list of iconic sweets, names, and recipes of which are closely allied to specific districts, such as Bogura, Cumilla, Tangail, Gopalganj, Natore, and Kushtia. These GI certifications regulate where the names of products are used, and aid in regional branding, but are vulnerable to counterfeiting and mislabeling in markets that sell non-GI products that visually resemble those provided under the certifications. In this case, automated visual recognition systems will contribute to safeguarding consumers as well as legitimate producers with the help of scalable, low-cost, dessert recognition and provenance validation tools. Existing Bangladeshi food image datasets, such as FoodBD and other traditional dessert and street-food collections offer good resources to generic classification tasks, but do not usually differentiate GI and non-GI products or capture explicit geographic provenance [2] [3]. In addition, these datasets are likely to group similar things visually in the same category which is not in line with the more rigorous differences required to authenticate GI. BD-GI-Desserts13 was therefore developed as a fine-grained, provenienced dataset which explicitly documents GI status and district labels, building upon and improving on existing resources [4] [5] [6] [7]. This dataset is also positioned within recent trends in deep learning food and dessert recognition, where the CCNN baseline trained in this work achieves 99.37% accuracy on BD-GI-Desserts13, demonstrating both the dataset’s difficulty and its potential as a benchmark for fine-grained recognition research.

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3 DATA DESCRIPTION

The BD-GI-Desserts13 dataset is a curated dataset of 13 GI-approved Bangladeshi dessert classes which are geographically specific. The dataset map Figure. 1 visually represents the spatial distribution of these classes throughout Bangladesh, with each marker indicating the main source district of a given dessert, which includes Bogura Doi, Natore Kacagolla, Kushtia Tiler Khaza, Tangail Chomchom and Sondesh variants, Sherpur Chanapayesh, Cumilla Rashmalai, Gopalganj Rosogolla. This number brings out clearly the connection between each dessert category and the area where the dessert is known in GI, and it is a direct mirror of the Location aspect of the Location dessert variant naming system of the dataset.

dataimages/Map.drawio.png

Figure 1: District-level geographic distribution of the 13 GI-certified dessert classes in BD-GI-Desserts13, showing the primary source district for each product in the BD-GI-Desserts13 dataset.

The acquisition of images of all 13 classes was only performed at the respective original GI-certified vendors in these areas. In each of the classes, the images were taken in-store in the actual stores and manufacturing facilities instead of being taken in second or third parties or online, meaning that each sample is a real product in its country

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dataimages/vendors.png

Figure 2: Representative on-site collection venues for three BD-GI-Desserts13 classes: (a) Cumilla Rashmalai, (b) Natore Kacagolla, and (c) Sherpur Chanapayesh.

of origin. As shown in the Figure. 2 representative examples of these vendors, such as the stores of Cumilla Rashmalai, Natore Kacagolla and Sherpur Chanapayesh, depicting the general trading conditions and shop fronts at which data were collected. All the pictures were taken in portrait mode and by use of iPhone devices in natural ambient lighting. The images obtained were then classified according to a hierarchical tree of folders as summarized in Figure. 3. At the highest tier, the repository is divided into three disjunctive subsets: train, val and test. The subdirectories of each subset are 13 class-specific subdirectories named according to the Location_dessert_variant scheme (e.g., Bogura_doi, Muktagacha_monda_brown, Tangail_sondesh_white). The prefix in this system represents the geographical provenance of the dessert, and the suffix, the type of the dessert and, where it exists, the visual or compositional variant. Some labeling errors, like additional spaces in Brahmanbaria chana muhki Misti and a small case prefix in kushtia tiler khaza, have been noted and will be fixed in later editions.

The total size of the dataset is 11,692 RGB images in JPEG format that take approximately 1 GB of storage space. All images were resized to 512 x 512 pixels to

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dataimages/dataset_folder.png

Figure 3: Hierarchical folder structure of BD-GI-Desserts13 following the Location_dessert[_variant] naming convention.

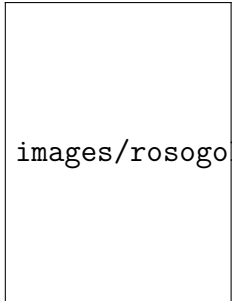
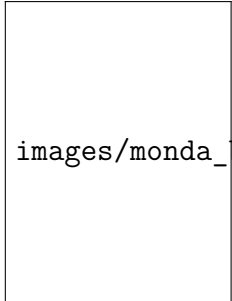
have a consistent input resolution across deep learning models. The corpus is relatively equal with each of the 13 classes having an average of 900 images. These data are divided into training, validation and test sets with a 70/15/15 proportion, which gives 8,183 training images, 1,752 validation images, and 1,757 test images; the number of images per class varies between 894 and 900 images as a result of curation and rounding. Augmented samples were created and stored with the original images in the same class directories to add more information to the dataset. Files with the suffix `_aug.jpg` represent standard geometric and photometric transformations (like rotation, flipping and brightness/contrast changes), whereas files with the suffix `_zoom.jpg` are zoom-based spatial transformations. An integrity check on the dataset-wide verified that all files can be read, that the files have the desired resolution, and that the files are present in a single split. The visual similarity variants, i.e. the brown and white versions of monda and sondesh, make BD-GI-Desserts13 specifically effective in recognizing fine-grained images with high inter-class similarity.

Table 1: Classes of traditional Bangladeshi sweets used in the dataset with descriptions and sample images.

Class Name	Description	Sample Image
Bogura Doi	Bogura Doi, also called Bograr Mishti Doi, is a thick, sweet yogurt from Bangladesh that has a creamy, custard-like texture. It’s made by slowly boiling cow’s milk and letting it set in traditional clay pots, which gives it a nice caramel-colored top and a balanced sweet-and-sour taste. It is a very popular dessert and even became an officially recognized regional product in 2023.	images/doi.png

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Table 1 continued from previous page

Class Name	Description	Sample Image
Brahmanbaria Chhanamukhi Mishti	Brahmanbaria Chhanamukhi Mishti is also known as Chhanamukhi is a famous chhena-based confection from Brahmanbaria, eastern Bangladesh. It has a hard, crystalline sugar coating and a sweet, moist dairy center. Made from fresh chhena shaped into small quadrilateral pieces, cooked in sugar syrup, and dried, it gains a unique crunchy-yet-soft texture and rich milky flavor. Carrying approximately two centuries of regional heritage, it was registered as a GI product in 2024.	 images/chanamukhimisti.png
Cumilla Rasmalai	Cumilla Rasmalai (also known as Cumillar Roshomalai) is a famous milk-based dessert from Cumilla, eastern Bangladesh. It has a soft, spongy texture and rich, sweet dairy flavor. Made from small chhena dumplings soaked in thick, slowly boiled milk (malai), it gains a velvety consistency and melts in the mouth. Registered as a GI product in 2024.	 images/rashmalai.png
Gopalganj Rosogolla	Gopalganj Rosogolla (also known as Gopalganjer Rosogolla) is a famous syrupy dessert from Gopalganj, central Bangladesh. It has a soft, spongy texture and rich, sweet flavor. Made from fresh chhena kneaded into smooth balls and boiled in light sugar syrup, it gains perfectly porous consistency and melt-in-the-mouth quality. Registered as a GI product in 2024.	 images/rosogolla.png
Muktagacha Monda Brown	Muktagacha Monda (also known as Gopal Paler Monda) is a famous traditional sweet from Muktagacha, Mymensingh, northern Bangladesh. Known for its rich brown color, it has a soft, slightly grainy texture and a deep, earthy sweet flavor. Made from fresh chhena (curd) slowly cooked with jaggery (gur) instead of sugar and shaped into small, flat patties, it gains a unique melt-in-the-mouth consistency and a warm, molasses-like aroma. Registered as a GI product in 2024.	 images/monda_brown.png

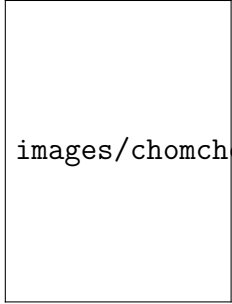
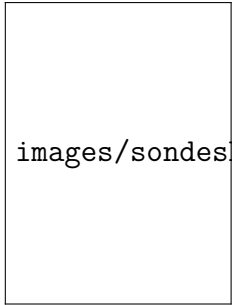
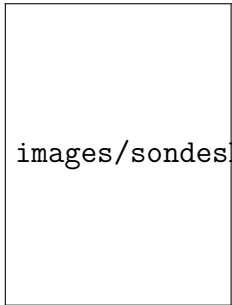
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Class Name	Description	Sample Image
Muktagacha Monda White	Muktagacha Monda (also known as Gopal Paler Monda) is a famous traditional sweet from Muktagacha, Mymensingh, northern Bangladesh. Known for its distinct white color, it has a soft, slightly grainy texture and a pure, rich dairy flavor. Made from fresh chhena (curd) slowly cooked with white sugar and shaped into small, flat patties, it gains a unique melt-in-the-mouth consistency and a delicate sweet aroma. Registered as a GI product in 2024.	images/monda_white.png
Natore Kachagolla	Natore Kachagolla (also known as Natore-er Kachagolla) is a famous chhena-based sweet from Natore, northern Bangladesh. It has a soft, crumbly texture and a mildly sweet dairy flavor. Made by mixing fresh raw chhena with sugar and lightly cooking it without forming it into tight shapes, it gains a unique granular consistency and rich taste. Registered as a GI product in 2023.	images/kachagolla.png
Netrokona Balish Mishti	Netrokona Balish Mishti is a famous Geographical Indication (GI) sweet from Netrokona, northern Bangladesh. Shaped like a soft pillow, it is made from pure chhana (cottage cheese), milk, and sugar. It features a delicate spongy texture with a rich creamy malai layer on top. This melt-in-the-mouth sweet is a beloved symbol of Netrokona's culinary heritage. Registered as a GI product in 2025.	images/balishmishti.png
Sherpur Chanapayesh	Sherpur Chhanar Payesh is a famous milk-based dessert from Sherpur, northern Bangladesh. It has a soft, granular texture and a rich, sweet dairy flavor with a hint of cardamom. Made by forming fresh chhena into tiny balls, soaking them in light sugar syrup, and simmering them in thick, slowly boiled milk (kheer), it gains a wonderfully creamy and melt-in-the-mouth consistency. Registered as a GI product in 2024.	images/chanapayesh.png

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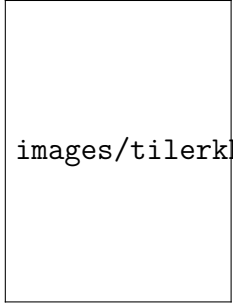
Table 1 continued from previous page

Class Name	Description	Sample Image
Tangail Chomchom	Tangail Chomchom (also known as Porabari Chomchom) is a famous traditional sweet from Porabari, Tangail, central Bangladesh. It has a firm, deeply caramelized outer crust and a soft, porous, syrup-drenched center. Made from fresh chhana slow-cooked in boiling sugar syrup and often rolled in dry mawa (crumbled milk solids), it gains a unique melt-in-the-mouth consistency and an intensely rich, sweet flavor. Registered as a GI product in 2024.	 images/chomchom.png
Tangail Sondesh Brown	Tangail Sondesh (also known as Tangailer Jamurkir Sandesh) is a famous traditional sweet from Jamurki, Mirzapur, Tangail, central Bangladesh. Known for its rich brown color, it has a soft, melt-in-the-mouth texture and a deep, earthy sweet flavor with the distinct aroma of date palm jaggery. Made from fresh chhena slowly cooked with patali gur (date jaggery) after boiling high-quality cow's milk for hours, it gains a unique velvety consistency and a highly praised, warm molasses-like aroma. Carrying a legacy from the British era, it was registered as a GI product in 2025.	 images/sondesh_brown.png
Tangail Sondesh White	Tangail Sondesh (also known as Tangailer Jamurkir Sandesh) is a famous traditional sweet from Jamurki, Mirzapur, Tangail, central Bangladesh. Known for its pristine white color, it has a soft, melt-in-the-mouth texture and a pure, rich dairy flavor with hints of cardamom. Made from fresh chhena slowly cooked with white sugar after boiling high-quality cow's milk for hours, it retains its bright white appearance while gaining a unique velvety consistency and a highly praised sweet aroma. Carrying a legacy from the British era, it was registered as a GI product in 2025.	 images/sondesh_white.png

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Table 1 continued from previous page

Class Name	Description	Sample Image
Kushtia Tiler Khaza	Kushtiar Tiler Khaja (also known as Tiler Khaja) is a famous traditional sweet snack from Kushtia, western Bangladesh. It has a crispy, flaky texture and a sweet, nutty sesame flavor. Made by boiling sugar or jaggery syrup, stretching it into thin layers, and coating it with roasted sesame seeds, it gains a uniquely crunchy consistency and delightful taste. Registered as a GI product in 2024.	 images/tilerkhaza.png

4 EXPERIMENTAL DESIGN, MATERIALS AND METHODS

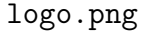
4.1 Experimental Design

The compilation of the BD-GI-Desserts13 dataset was done in a five-stage pipeline as illustrated in Figure. 4. First, dessert products were found at their original geographical areas in Bangladesh, such as Tangail, Sherpur, Muktagacha, Netrokona, Bogura, Natore, Kushtia, Brahmanbaria, Gopalganj and Cumilla. Raw images were taken at the site with the help of original vendors and producers with the use of Apple iPhone 13, iPhone 16, and iPhone 16 Pro Max in portrait mode. The images captured were then copied out of the device storage into a centralized local database and arranged into a hierarchical directory system on a class basis in accordance with the `Location_dessert_variant` naming system. In this stage, images were investigated by hand and low-quality images (blurred, over- or underexposed, duplicated or class-ambiguous) were eliminated.

dataimages/method.png

Figure 4: Five-stage data collection and curation pipeline for BD-GI-Desserts13.

The curated image corpus was then processed in the last stage of the pipeline by a single preprocessing and augmentation procedure that generated three dataset variants. The



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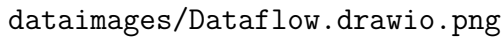
Original Images version only has the samples collected in the field filtered out. The Resized Images version includes all of the images that have been scaled to a 512×512 -pixels size in RGB color space and are in JPEG format with the EXIF metadata stripped. The Augmented Images version consists of the extra samples created through geometric and photometric transformations (horizontal flipping, rotation, brightness and contrast adjustment, and zoom-based cropping) stored as files with the suffixes of `_aug.jpg` and `_zoom.jpg`, and their original counterparts in the training split.

4.2 Materials (Camera Specification)

The pictures in the BD-GI-Desserts13 dataset were all taken using three Apple iPhone models namely: iPhone 13, iPhone 16, and iPhone 16 Pro Max. All the devices were used through the default camera app in portrait (mobile) mode during the process of data collection. The iPhone 13 features a 12 MP wide camera with $2\times$ optical zoom and $f/1.6$ aperture; the iPhone 16 features a 48 MP wide camera with $2\times$ optical zoom and $f/1.6$ aperture; and the iPhone 16 Pro Max features a 48 MP wide camera with $5\times$ optical zoom and advanced optical image stabilization. The devices all generated JPEG in the RGB color space. No external lenses, tripods, artificial lighting, or other photographic accessories were used; flash was turned off on all sessions and no photographic adjustments to the camera (like HDR overrides, portrait-depth effects and digital zoom) were made.

4.3 Methods

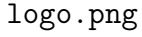
The BD-GI-Desserts13 data processing pipeline starts with multi-location field acquisition and finishes with three formatted dataset variants as shown by the Figure. 5. All the images were copied to a centralized local database after they were captured on-site. Every image underwent a visual screening process, with only images of quality and class-correctness retained; those that were degraded by motion blur, grossly over- or under-exposed, duplicated or unidentifiable by their class. The other images were sorted into a class-based folder structure based on the `Location_dessert_variant` naming scheme with the same class subdirectories duplicated at the `train/`, `val/`, and `test/` root.



dataimages/Dataflow.drawio.png

Figure 5: End-to-end data processing pipeline from field acquisition to three formatted dataset variants.

The Pillow library was then used on Python 3.x to perform a standardized preprocessing pipeline. All pictures were resized to 512×512 pixels using bicubic interpolation, tested or transformed to RGB color space and saved as `.jpg` files with all embedded EXIF metadata removed. Later data augmentation was limited to the

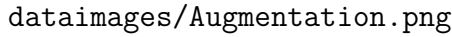


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`train/` split, generating two groups of augmented variants: standard augmentations (horizontal flipping, random rotation, and brightness/contrast adjustment) and zoom spatial augmentations. The augmented images were archived in the same class directories as the origin images with identical suffixes on the filenames. An integrity audit after processing, with custom python scripts, confirmed that all 11,692 files were readable, with the appropriate resolution and showed no cross-partition overlap of files.

4.4 Data Augmentation

Data augmentation was done offline to enhance the variability within a single class and to enhance the robustness of the model, as indicated in the Figure. 6. All changes were done in python with Pillow library and only training images had changes applied; validation and test sets were not modified. To rotate, a random rotation of the source image was performed within an angular range of about ± 18 –20 degrees about the center of the source image without distorting the aspect ratio. Brightness was adjusted by changing the pixel intensities to make the images appear about 10–15% darker or brighter than the original, and contrast was adjusted by multiplicative factors that reflected about 10–13 percent change in contrast. Other changes were random zooming (in and out) to simulate camera distance changes, the addition of subtle grain-like noise to simulate sensor noise and low-light situations, and random cropping and resizing to 512×512 pixels to simulate different framings. All augmented files were stored in the same class directories as the originals with consistent naming suffixes to differentiate between augmented and field-collected samples.

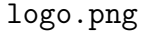


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dataimages/Augmentation.png
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Figure 6: Offline data augmentation pipeline applied to the training split of BD-GI-Desserts13.

4.5 Model Validation

The proposed Custom Convolutional Neural Network (Custom CNN) is a compact deep learning architecture designed to achieve an optimal balance between classification accuracy and computational efficiency [8] [9]. The model can efficiently capture hierarchical visual features with minimal computational costs through stacked convolutional architecture with batch normalization and pooling operation. These properties make the Custom CNN an appropriate backbone to the automated identification of Bangladeshi GI dessert classes, where subtle visual differences in the



texture, colour, and shape pose a challenging classification task, and where scalable inference at speed is a feasible requirement [10] [11].

The proposed Custom Convolutional Neural Network (CCNN) is a lightweight deep learning architecture designed to achieve an optimal balance between classification accuracy and computational efficiency. All input images are resized to 224×224 RGB format before being fed into the model, regardless of their original storage resolution of 512×512 . The overall architecture of the proposed CCNN is illustrated in Figure. 7. The model captures hierarchical visual representations through stacked convolutional blocks incorporating batch normalization and max-pooling, ensuring both effective feature extraction and reduced computational cost. This design makes the CCNN suitable for automated classification of Bangladeshi GI dessert classes, where fine-grained recognition is challenging due to subtle variations in texture, color, and shape. The CCNN consists of five convolutional blocks, each containing two consecutive Conv2D layers (with 3×3 kernels, same padding, and ReLU activation), followed by Batch Normalization and a MaxPooling2D layer (pool size 2×2). The number of filters increases progressively across the blocks: 32, 64, 128, 256, and 512. Through five successive 2×2 pooling operations, the spatial resolution is reduced from 224×224 to 7×7 , resulting in a feature volume of $7 \times 7 \times 512$ at the end of the fifth block. This feature map is then flattened into a one-dimensional vector of size 25,088 ($7 \times 7 \times 512 = 25,088$). A fully connected Dense layer with 500 neurons and ReLU activation follows, after which a Dropout layer (rate = 0.40) is applied for regularization. Finally, a Softmax output layer with 13 neurons is used for classification, corresponding to the 13 GI dessert categories. The model comprises a total of 17,271,173 parameters, of which 17,267,205 are trainable (approximately 17.27 million parameters).

Training was conducted for 30 epochs using a batch size of 32 and the Adam optimizer with an initial learning rate of 5×10^{-4} . To ensure stable convergence and prevent overfitting, ReduceLROnPlateau and EarlyStopping callbacks were employed.

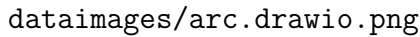


Figure 7: Architecture of the proposed Custom CNN (CCNN) showing five convolutional blocks and the fully connected classification head.

Table 2: Class-wise image counts and train/validation/test split distribution in BD-GI-Desserts13.

Class	Original Image	Augmented Image	Train	Val	Test
Bogura_doi	312	900	630	135	135
Brahmanbaria_chana_mukhi_misti	310	900	630	135	135
Cumilla_rashmalai	395	900	630	135	135
Gopalganj_rosogolla	333	900	630	135	135
Muktagacha_monda_brown	326	899	629	134	136
Muktagacha_monda_white	301	900	630	135	135
Natore_kacagolla	301	894	625	134	135
Netrokona_balish_mishti	341	900	630	135	135
Sherpur_chanapayesh	372	899	629	134	136
Tangail_chomchom	331	900	630	135	135
Tangail_sondesh_brown	456	900	630	135	135
Tangail_sondesh_white	356	900	630	135	135
Kushtia_tiler_khaza	303	900	630	135	135
Total	4137	11692	8183	1752	1757

Custom Convolutional Neural Network (CCNN) was used to predict the GI-certified categories of desserts in Bangladesh on a multi-class image dataset of both visually similar and dissimilar dessert samples obtained in realistic, unconstrained field conditions across a number of districts. The dataset composition and the number of original and augmented images per class is outlined in the Statistics of the Dessert Dataset Table. 2, and it is clear that all thirteen categories of desserts are sufficiently represented in order to support the supervised learning process. To improve the generalization performance of the model in relation to changes in illumination, viewpoint, background and capture environment, the data augmentation was done solely to the training subset, but the validation set was left in its original form.

$$Accuracy = \frac{\text{Number of Correct Predictions}}{\text{Total Number of Predictions}} \quad (1)$$

$$Precision = \frac{\text{True Positive}}{\text{True Positive} + \text{False Positive}} \quad (2)$$

$$Recall = \frac{\text{True Positive}}{\text{True Positive} + \text{False Negative}} \quad (3)$$

$$F_1 = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (4)$$

The dataset was divided into two training and validation groups, with a fixed proportion split of both subsets, so that every dessert category is represented on both subsets proportionately to reduce the effects of class imbalance. Custom CNN was trained with mini-batches stochastic optimization with a fixed batch size and a dynamic learning rate schedule, which slowly decreases learning rate given training time to ensure stable convergence. The training subset was the only one that was online

logo.png

dataimages/confusion_matrix.png

Figure 8: Per-class confusion matrix of the CCNN evaluated on the BD-GI-Desserts13 test set.

augmented during training, with the validation subset used to monitor the performance and provide a final evaluation of the model. The classification performance of the proposed model was quantitatively assessed using accuracy, precision, recall, and F_1 -score, computed from the numbers of true positives, true negatives, false positives, and false negatives for each class.

According to Table. 3 the per-class and overall classification performance, including precision, recall, the F_1 -score and overall accuracy, indicate that the proposed Custom CNN is highly discriminative in all thirteen GI dessert categories. Most of the classes have high accuracy and high recall rates, which indicates that the model can not only recognise each class of desserts, but also has a relatively low error rate, which signifies its strength in the context of fine-grained similarities between GI-certified dessert classes, such as similarities in colour and texture between brown and white variations of Muktagacha Monda and Tangail Sondesh. In order to explore the error structure of the classifier further, a confusion matrix was constructed on the held-out test set and is shown in Figure. 8. The matrix gives a more detailed description of the correct prediction (principal diagonal terms) and inter-class confusion (off-diagonal terms), thus showing which particular categories of desserts are most subject to misclassification.

Table 3: Per-class precision, recall, and F_1 -score of the CCNN baseline on the BD-GI-Desserts13 test set.

Class Name	Precision	Recall	F1-score
Bogura_doi	1.0000	0.9333	0.9655
Brahmanbaria_chana_mukhi_misti	1.0000	0.9852	0.9925
Cumilla_rashmalai	1.0000	1.0000	1.0000
Gopalganj_rosogolla	1.0000	1.0000	1.0000
Muktagacha_monda_brown	1.0000	1.0000	1.0000
Muktagacha_monda_white	1.0000	1.0000	1.0000
Natore_kacagolla	1.0000	1.0000	1.0000
Netrokona_balish_mishti	1.0000	1.0000	1.0000
Sherpur_chanapayesh	1.0000	1.0000	1.0000
Tangail_chomchom	1.0000	1.0000	1.0000
Tangail_sondesh_brown	1.0000	1.0000	1.0000
Tangail_sondesh_white	0.9854	1.0000	0.9926
Kushtia_tiler_khaza	0.9375	1.0000	0.9677
Macro Avg	0.9941	0.9937	0.9937
Weighted Avg	0.9941	0.9937	0.9937
Overall Accuracy			99.37%

This type of analysis may be particularly helpful in identifying systematic error patterns, like confusion of desserts with similar chhena appearance, syrup-covered surface, and may also serve as the basis of analysis in the future in data acquisition strategies, class boundaries, or architectural refinements of the proposed model. Also, the dynamics of training of the Custom CNN model were analyzed by evolving the accuracy and loss on the basis of epochs as illustrated in the Accuracy and loss graph of Custom Convolutional Neural Network (CCNN) Figure. 9. The training and validation accuracy curves illustrate the steady increase in the recognition performance, with the loss curves illustrating the steady loss reduction and ultimate stagnation, which indicates successful optimization and lack of gross overfitting. The fact that training and validation metrics approach convergence across epochs is an indication that the last Custom CNN setting has a desirable trade-off between model capacity and regularization, resulting in good generalization to unseen samples.

Overall, BD-GI-Desserts13 dataset has been assembled in a five-stage pipeline that involved the field collection, quality filters by hand, class-based organisation, standardised preprocessing, and controlled augmentation. Under ambient conditions, ten districts were photographed with three consumer-grade smartphones, returning 11,692 integrity-verified RGB images of 512x512 pixels, of thirteen GI-certified Bangladeshi desserts. A class-stratified 70/15/15 split is used to ensure balanced

logo.png

dataimages/training_curves.png

Figure 9: Training and validation accuracy (right) and loss (left) curves of the CCNN over 30 epochs on BD-GI-Desserts13.

representation across all categories. Both the discriminative power of the dataset and the stability of convergence and the absence of gross overfitting were validated, which reached a test accuracy of 99.37 percent and a stable converged value without gross overfitting, demonstrating BD-GI-Desserts13 as a trusted benchmark dataset for fine-grained GI dessert recognition.

5 LIMITATIONS

While BD-GI-Desserts13 is relatively large for a specialized GI dessert dataset, with 11,692 images across thirteen classes, it remains smaller than some large-scale generic food datasets, which may limit the training of very deep architectures without transfer learning. The images were collected from a finite number of shops and households in selected districts, so the dataset may not capture every stylistic or seasonal variation of each GI dessert throughout Bangladesh.

Moreover, the dataset contains currently only image-level class labels and rough metadata on district and capture environment. It does not have bounding boxes on various items, pixel-wise segmentation or ingredient tags, and restricts certain forms of analysis such as accurate portion estimation or multi-object detection. Finally, the dataset indicates the standpoint of the Bangladeshi GI registry at the time of collecting; additional GI desserts to be put into practice later are not covered by it, and an extension will be needed in the future to get it updated to correspond with the policy changes.

ETHICS STATEMENT

There are no human participants, animals, and social-media data experiments involved in this work. The photos were taken in open and business places like sweet shops, markets and in homes with verbal consent of the owners or hosts of the shops and there are no recognizable human faces and personal details as well.

DATA AVAILABILITY

The dataset used in this study (BD-GI-Desserts13) is publicly available and can be accessed at the following link:[BD-GI-Desserts13 Dataset Repository](#)

CRedit AUTHOR STATEMENT

Shakibur Rahman Bhuiyan: Methodology, Data curation, Investigation, Writing – review & editing. **Dip Kundu:** Data curation, Methodology, Writing – review & editing, Writing - original draft. **MD Tamzed Hossain Siam:** Data curation, Formal analysis, Visualization. **Zarif Wasif Bhuiyan:** Supervision, Validation, Formal analysis, Visualization. **Asma Begum:** Investigation, Formal analysis, Validation. **Md Tarek Habib:** Conceptualization, Supervision, Project administration, Resources.

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DECLARATION OF COMPETING INTERESTS

The authors indicate that they do not have any known competing financial interests or personal relationship that may have emerged to affect the work reported in this article.

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